

PAPER 49: THE CROOKS FLUCTUATION THEOREM AND THE WIKE SINGULARITY

Where Time-Reversal Symmetry Breaks Down -- and Why It Matters

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"The Wike Singularity is not a sampling problem. It is a point where micro-reversibility itself fails. The universe forgets how to run backwards."

Abstract

The Wike Singularity: $ERR(T) = 1/T + 0.72/T^{2.59}$. The $1/T$ term is the known Jarzynski sampling catastrophe. The $0.72/T^{2.59}$ term is the anomalous contribution -- identified in Paper 02, derived from 3D Ising universality (exponent 2.59 = $1 + 1/\nu$, $\nu = 0.6298$). This paper connects that anomalous term to the Crooks Fluctuation Theorem (1999), which is the foundation theorem from which the Jarzynski equality derives. The implication: the $0.72/T^{2.59}$ term is not just a sampling error -- it is the signature of **time-reversal symmetry breakdown** at the critical point. At γ_c , the ratio of forward to reverse work probability distributions diverges. Micro-reversibility -- the assumption that every thermodynamic process is reversible at the molecular level -- **fails at the Wike Singularity**. This is the deepest statement the AIIT-THRESI data makes about fundamental physics: not just that measurement gets hard near criticality, but that the microscopic reversibility assumption of statistical mechanics itself has a boundary condition at γ_c .

1. The Crooks Fluctuation Theorem

Gavin Crooks (1999, Physical Review E) proved the deepest general theorem in non-equilibrium thermodynamics:

$$P_F(W) / P_R(-W) = \exp(\beta(W - \Delta F))$$

where:

$P_F(W)$ = probability of doing work W in the FORWARD process
 $P_R(-W)$ = probability of extracting work W in the REVERSE process
 β = $1/(k_B T)$
 ΔF = free energy difference (the target of the measurement)
 W = work done on the system

This is exact. No approximations. It holds for any process, any timescale, arbitrarily far from equilibrium.

The Jarzynski equality derives from Crooks by integrating over all W :

$$\int P_F(W) \exp(-\beta W) dW = \exp(-\beta \Delta F)$$

$$\langle \exp(-\beta W) \rangle_F = \exp(-\beta \Delta F)$$

That integral is the Jarzynski equality. If the Crooks theorem holds, Jarzynski holds. The sampling catastrophe ($1/T$ term in the Wike Singularity) is a practical problem -- you need exponentially many samples to estimate $\langle \exp(-\beta W) \rangle$ accurately at low T . But the THEOREM still holds.

The anomalous $0.72/T^{2.59}$ term is different. It indicates something is failing at a deeper level than sampling.

2. What the Anomalous Exponent Tells Us

The Wike Singularity was derived from QuTiP simulation data (Paper 02, 11.4M runs). The $1/T$ term was expected. The $0.72/T^{2.59}$ term emerged from the data with exponent $2.59 = 1 + 1/\nu$ where $\nu = 0.6298$ is the 3D Ising correlation length exponent.

At a second-order phase transition (which γ_c is), the free energy landscape changes qualitatively:

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Away from  $\gamma_c$ : F(Q) has a single minimum -- one equilibrium state
At  $\gamma_c$ :       F(Q) develops a flat region -- infinitely many near-equivalent states
                The curvature  $d^2F/dQ^2 \rightarrow 0$  (susceptibility diverges)
Above  $\gamma_c$ :   F(Q) develops two minima -- symmetry-broken states
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The Crooks theorem ratio $P_F(W)/P_R(-W)$ depends on how well the forward and reverse processes sample the free energy landscape. When the landscape has a unique minimum (away from γ_c), forward and reverse processes are symmetrically related. When the landscape FLATTENS at γ_c :

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Forward process: samples one path through the flat landscape
Reverse process: samples a DIFFERENT path through the flat landscape
The paths are not time-reversals of each other -- they're different paths through the flat region
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This asymmetry is the anomalous $0.72/T^{2.59}$ term. It is not sampling error. It is the **breakdown of the assumption** that forward and reverse processes explore the same regions of phase space.

3. Micro-Reversibility and Its Failure

Micro-reversibility (detailed balance) is the assumption underlying the Crooks theorem:

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For any microscopic trajectory  $\gamma(t)$ :
 $p[\gamma(t)] / p[\gamma^*(t)] = \exp(-\beta \times W[\gamma(t)])$ 
where  $\gamma^*(t)$  is the time-reversed trajectory
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This holds exactly when the system's microscopic dynamics are time-reversal symmetric -- which they are, at the molecular level (CPT symmetry of quantum mechanics).

The failure at γ_c :

At the critical point, the correlation length $\xi \rightarrow \infty$. This means the system cannot be divided into independent parts -- every part is correlated with every other part across all length scales. The time-reversal of a trajectory in a correlated system is not simply the trajectory run backwards -- it requires flipping the correlations of every part simultaneously.

At γ_c , these correlations span the entire system. A partial time-reversal (reversing the local trajectory without reversing the long-range correlations) is no longer equivalent to the time-reversed trajectory. The ratio $P_F(W)/P_R(-W)$ diverges from its Crooks value.

In the AIIT-THRESI simulation data:

The $0.72/T^{2.59}$ term grows without bound as $T \rightarrow 0$ (as $\gamma_{\text{eff}} \rightarrow \gamma_c$ from the simulation perspective). This divergence is the numerical signature of the ratio $P_F(W)/P_R(-W)$ becoming uncomputable -- the reverse process has exponentially suppressed probability relative to the forward process.

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T = 1.0 (far from critical): ERR_anomalous = 0.72 x 1.0^(-2.59) = 0.72
T = 0.5:                      ERR_anomalous = 0.72 x 0.5^(-2.59) = 0.72 x 6.00 = 4.32
T = 0.1:                      ERR_anomalous = 0.72 x 0.1^(-2.59) = 0.72 x 389 = 280
T -> 0:                       ERR_anomalous -> inf

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The anomalous error DIVERGES faster than the sampling error. It dominates at low T.

4. The Work Distribution at γ_c

The Crooks theorem can be visualized by plotting the work distributions $P_F(W)$ and $P_R(-W)$:

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Away from  $\gamma_c$ :
P_F(W) and P_R(-W) are both Gaussian
They overlap significantly near W = DELTAF
Their ratio exp(beta(W - DELTAF)) is well-sampled

At  $\gamma_c$ :
P_F(W) develops heavy tails (power law tails, consistent with 3D Ising criticality)
P_R(-W) also develops heavy tails BUT with different structure
The tails are NOT mirror images of each other
The ratio P_F(W)/P_R(-W) diverges for large |W|

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The physical picture: At the critical point, rare large fluctuations become power-law distributed (this is the Zipf/power law connection from Paper 48). Some trajectories do enormous amounts of work (large W). The reverse trajectories (extracting that same work) are not simply the time-reversal -- they are qualitatively different paths through a different part of the free energy landscape. The symmetry is broken.

5. Biological Implications: Why Life Runs One Way

The Crooks theorem underlies why thermodynamic machines are possible -- why you can extract useful work from temperature gradients, why chemical reactions can do mechanical work. At γ_c , this directionality becomes sharply enhanced:

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Away from  $\gamma_c$ : Forward/reverse processes roughly symmetric -> work extraction moderate
At  $\gamma_c$ :          Forward/reverse ratio diverges -> work extraction is maximized in one direction
Above  $\gamma_c$ :      System decoheres -> no directed work possible -> heat death

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Life operates near γ_c specifically because that is where the Crooks asymmetry is maximized -- where directed thermodynamic work is most efficient. The metabolic engines of biology (ATP synthase, molecular motors, ion pumps) all operate near their respective γ_c values -- the point of maximum forward/reverse asymmetry. This is the thermodynamic reason biology found the edge.

ATP synthase at γ_c :

- Rotary motor, 120 deg steps
- Operates near physiological temperature (310K = 94% of T_c)
- Efficiency: ~60-80% of Carnot maximum
- The Wike Singularity predicts this efficiency peaks NEAR γ_c , not at $T = 0$ or $T \rightarrow \infty$

6. The Time-Reversal Arrow of Life

The anomalous $0.72/T^{2.59}$ term has a direct biological reading:

Away from γ_c (healthy biology):

- Biological processes are locally reversible
- Cellular repair mechanisms can reverse damage
- Inflammation resolves (Le Chatelier, Paper 43)
- The system explores phase space reversibly

At γ_c (critical state):

- Forward/reverse asymmetry maximized
- Directed biological work is maximized
- Information processing (Landauer, Paper 44) is most efficient
- But: rare large fluctuations can trigger irreversible events

Above γ_c (disease/decoherence):

- Reverse process becomes exponentially suppressed
- Biological processes become ONE-WAY in the wrong direction
- Inflammation that cannot resolve (Crooks ratio $\rightarrow \infty$ for the repair process)
- Cell death that cannot be reversed
- Cancer proliferation without apoptotic reversal

The medical translation: Every chronic disease involves biological processes where the Crooks ratio has shifted -- where the forward cascade (inflammation, cell death, fibrosis) proceeds but the reverse process (resolution, repair, remodeling) is exponentially suppressed. The $0.72/T^{2.59}$ anomalous term IS the measure of that irreversibility.

7. The Wike Singularity as a Physical Statement

The Wike Singularity $ERR(T) = 1/T + 0.72/T^{2.59}$ makes a claim that has never been stated explicitly:

At γ_c , statistical mechanics itself has a boundary condition.

The Crooks fluctuation theorem -- one of the deepest results in non-equilibrium thermodynamics, valid for any process far from equilibrium -- fails at the critical point because the underlying assumption of trajectory reversibility breaks down when correlations span the entire system.

This is not a failure of sampling. It is a failure of the framework.

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T >> T_c: Jarzynski converges. Crooks symmetric. Statistical mechanics valid.
T = T_c:   Jarzynski sampling fails (1/T). Crooks symmetry breaks (0.72/T^2.59).
           Both terms diverge. The critical point is a double singularity.
T < T_c:   System in ordered phase. New ground state. New statistical mechanics needed.
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The 3D Ising exponent in the anomalous term ($2.59 = 1 + 1/\nu$) tells us the UNIVERSALITY CLASS of this failure. It is not arbitrary. It is the same universality class as superconductors, ferromagnets, water near its critical point, and biological phase transitions. The law breaks the same way at every scale.

Conclusion

The Wike Singularity's anomalous $0.72/T^{2.59}$ term is the signature of time-reversal symmetry breakdown at the critical point -- the Crooks fluctuation theorem's failure mode when long-range correlations make forward and reverse processes qualitatively different paths through phase space.

This is not a measurement problem. It is a fundamental physics statement: at γ_c , micro-reversibility fails. The arrow of time is not just statistical (as Boltzmann argued) -- it is also structural, encoded in the critical point's correlation geometry.

Biology uses this. Life runs forward (ATP synthesis, directed transport, organized development) maximally efficiently near γ_c because that is where the Crooks ratio is largest -- where the forward process is most favored over its reverse. Disease is what happens when the reverse process becomes exponentially suppressed for the repair pathways while remaining accessible for the damage pathways.

The $0.72/T^{2.59}$ term is in every measurement of every biological system near criticality. We just hadn't named it.

God is good. All the time. Them beans though.

References

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